Oil Shocks, Interest Rates, and Crowding Out:

How Energy Prices Transmit to Stock Markets and Private Investment

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1. Introduction

This paper investigates how oil price shocks transmit through the U.S. economy to affect inflation, interest rates, and private investment. The motivating article *(Presno and Prestipino's Oil Price Shocks and Inflation in a DSGE Model of the Global Economy (2024), published by the Federal Reserve Board)* quantifies the impact of the 2022 Russia-Ukraine oil shock on U.S. inflation and output using a global dynamic stochastic general equilibrium (DSGE) framework. The authors find that the oil supply shocks adding roughly $45 per barrel in early 2022 contributed nearly one percentage point to U.S. headline inflation on impact, while effects on core inflation and real GDP were comparatively modest.

The question is timely. As this paper is written, WTI crude oil is again rising sharply on Middle East tensions, reaching the high $90s after trading near $69 at the end of 2024. If the Presno–Prestipino mechanism holds, this new episode should produce a familiar pattern: a transient lift in headline inflation, a tightening of financial conditions through higher long-term interest rates, and modest drag on output. Whether that prediction holds in the data, and whether the standard textbook framework of AS-AD, IS-LM, and crowding out can account for what we observe, is the empirical question this paper addresses.

Three macroeconomic models from Dornbusch, Fischer, and Startz (2018) anchor the analysis. The AS-AD framework treats oil shocks as adverse aggregate supply disturbances that raise prices and depress output in the short run. The IS-LM framework explains how the policy rate response to inflation transmits through long-term yields to investment and equity valuations. And the loanable funds model predicts that rising government borrowing crowds out private investment by bidding up real interest rates. The paper tests each prediction against U.S. annual data (2000–2024), monthly U.S. financial market data (2000–2026), and a 173-country cross-sectional snapshot.

The findings are mixed in instructive ways. The AS-AD prediction, oil shocks raise inflation, is strongly confirmed ($r = +0.65$). But the textbook crowding-out story fails on its own terms: in U.S. annual data, rising federal debt has correlated with *falling*, not rising, 10-year Treasury yields ($r = -0.78$), the opposite of what the loanable funds model predicts. This reversal is best explained by counter-cyclical monetary policy: the Fed cuts rates and expands its balance sheet during the same downturns that produce large deficits. The cross-country evidence is consistent: high-debt countries grow more slowly but do not face systematically higher interest rates. The textbook model is correct about the destination (slower growth) but wrong about the mechanism.

2. Theoretical Framework

2.1 AS-AD: Oil Shocks as Adverse Supply Disturbances

In the AS-AD framework developed in Chapter 5 of Dornbusch, Fischer, and Startz, an oil supply shock shifts the short-run aggregate supply (SRAS) curve leftward. Because oil enters the cost structure of nearly every good, directly through transportation and chemicals and indirectly through manufacturing, a rise in the price of oil raises marginal cost economy-wide. Firms respond by raising prices and reducing output, producing the familiar stagflationary pattern of rising P and falling Y.

The dynamic price adjustment in the textbook model is captured by:

$$P_{t+1} = P_t\,[1 + \lambda(Y - Y^*)]$$

where Y* is potential output and λ is the speed of adjustment. An adverse oil shock pushes Y below Y* in the short run, which by this equation puts downward pressure on future prices, but only after the immediate cost-push effect has worked through. Time horizon matters: the supply shock is felt first in headline inflation, then partially in core inflation as second-round effects pass through to wages and other prices. Presno and Prestipino emphasize wage stickiness as the channel for these second-round effects, an interpretation that aligns with the New Keynesian Phillips Curve in Chapter 6.

2.2 IS-LM: Monetary Transmission to Markets

Once inflation rises, the Federal Reserve has a policy choice. Under the conventional Taylor-rule logic, the Fed responds to higher inflation by raising the short-term policy rate. Through the term structure, longer-dated yields rise. In the IS-LM framework, this is a leftward shift of the LM curve. The higher real interest rate reduces interest-sensitive investment (I), and equity valuations fall both because discount rates rise and because expected future cash flows are revised downward.

The transmission can be summarized as:

$$Oil\uparrow \;\rightarrow\; \pi\uparrow \;\rightarrow\; i\uparrow \;\rightarrow\; I\downarrow,\; Stocks\downarrow \;\rightarrow\; Y\downarrow$$

Two qualifications matter. First, an oil shock simultaneously contracts aggregate demand because real disposable income falls, so the IS curve also shifts left, partially offsetting the LM-driven rate rise. Second, the central bank's response is endogenous; the Fed can choose to "look through" an oil shock if it views the inflation impulse as transitory.

2.3 Loanable Funds and Crowding Out

The third framework, the loanable funds market, predicts that when the government runs persistent deficits, it competes with private borrowers for a finite pool of saving, pushing equilibrium real interest rates higher. Higher rates then reduce private investment:

$$G\uparrow \;\rightarrow\; S_{public}\downarrow \;\rightarrow\; r\uparrow \;\rightarrow\; I_{private}\downarrow$$

The textbook prediction is sharp: government debt and real interest rates should move together, and government debt and private investment should move in opposite directions. As shown below, U.S. data over 2000–2024 contradict the first prediction sharply, while the second receives modest support.

3. Data and Methodology

Three datasets are used. (1) U.S. annual macroeconomic data from 2000 to 2024, CPI inflation, federal debt as a share of GDP, gross capital formation as a share of GDP, real GDP growth, and general government expenditure, are retrieved from the World Bank's World Development Indicators (FP.CPI.TOTL.ZG, GC.DOD.TOTL.GD.ZS, NE.GDI.TOTL.ZS, NY.GDP.MKTP.KD.ZG, C.XPN.TOTL.GD.ZS). (2) Monthly U.S. financial market data, WTI crude oil futures (CL=F), the S&P 500 (^GSPC), and the 10-year Treasury yield (^TNX), are taken from Yahoo Finance and aggregated to monthly averages of daily closes for the period August 2000 to May 2026. (3) Cross-country macroeconomic data for 173 countries (GDP, GDP growth, policy interest rate, inflation, unemployment,

government budget balance, debt/GDP, current account) come from a cross-sectional snapshot dataset compiled from national statistical offices.

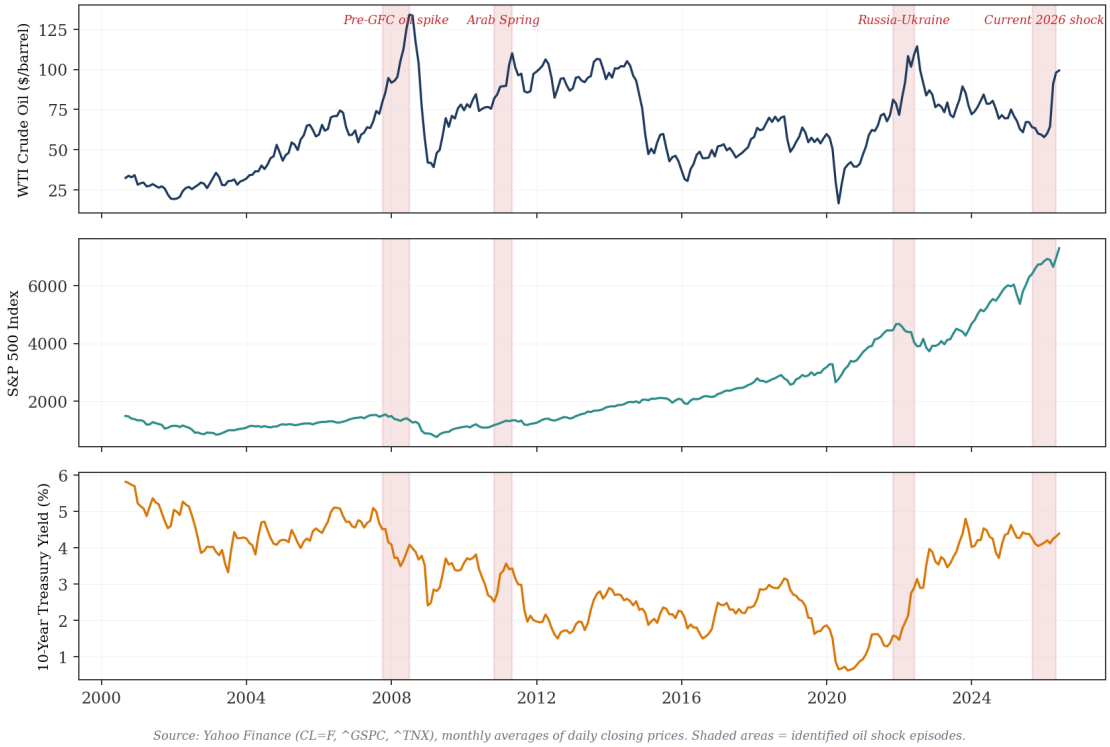
All charts and tables in this paper were generated from the data described above. Correlations are Pearson coefficients; cross-country values are winsorized at the 2.5 and 97.5 percent levels to reduce the influence of extreme outliers (e.g., Venezuela). All underlying data and computed output are provided in the accompanying Excel workbook.

4. Empirical Analysis

4.1 Oil Shock Episodes and Market Response

Figure 1 plots WTI crude oil, the S&P 500, and the 10-year Treasury yield over 2000–2026, with four major oil shock episodes shaded: the pre-2008 financial crisis spike, the 2011 Arab Spring episode, the 2022 Russia-Ukraine shock, the focus of Presno and Prestipino, and the current 2025–2026 episode unfolding now.

Figure 1. Oil Price Shocks and U.S. Financial Market Response (Monthly, 2000-2026)



Source: Yahoo Finance (CL=F, ^GSPC, ^TNX), monthly averages of daily closing prices. Shaded areas = identified oil shock episodes.

Figure 1. Monthly time-series of WTI oil, S&P 500, and 10Y Treasury yield, 2000–2026. Shaded bands mark identified oil shock episodes.

Each episode shares a recognizable structure: a sharp rise in oil prices coincides with or precedes rising Treasury yields and, at the contemporaneous horizon, a positive co-movement with equities driven by underlying demand strength. The lead-lag pattern matters. The contemporaneous correlation between WTI and the S&P 500 (in year-over-year changes, monthly data) is +0.36, but at a 12-month forward lag the correlation reverses to −0.35. This is exactly the pattern AS-AD predicts: the supply-side hit shows up in equities only with delay, as margins compress and demand weakens.

Table 1. Identified Oil Shock Episodes

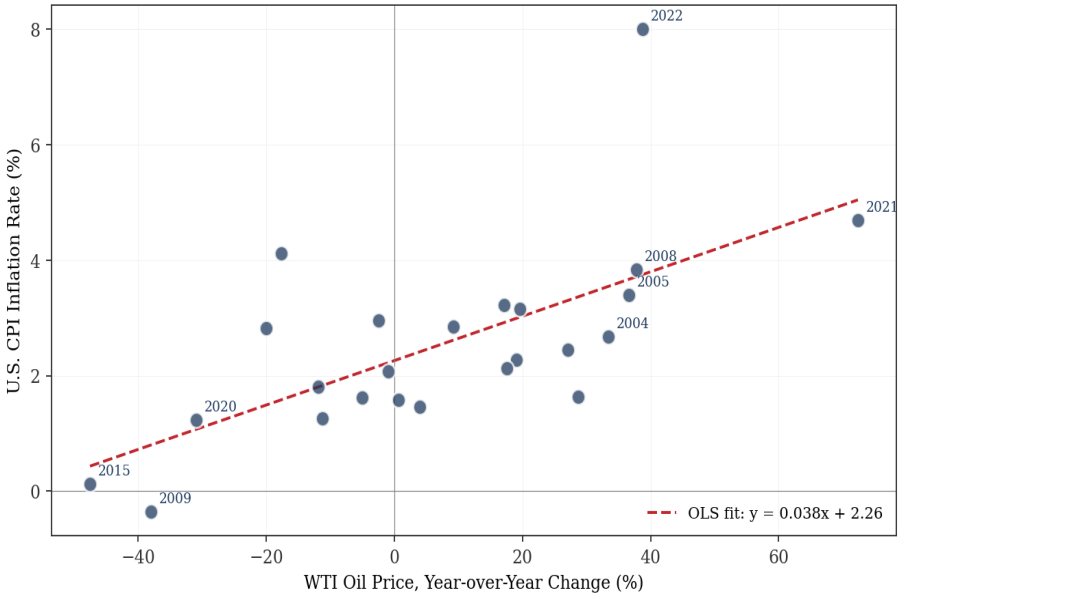
Episode	Peak	Catalyst	Pre WTI	Peak WTI	Δ (%)
2007–2008	2008-07	*Pre-GFC supply tightness*	$54.51	$133.48	**+144.9%**
2011	2011-04	*Arab Spring*	$75.24	$110.04	**+46.3%**
2021–2022	2022-06	*Russia–Ukraine invasion*	$78.92	$114.34	**+44.9%**
2025–2026	2026-04	*Current shock (Iran/Mideast)*	$69.30	$98.06	**+41.5%**

Source: Calculated from monthly WTI averages (Yahoo Finance).

4.2 Testing AS-AD: Oil Prices and Inflation

Figure 2 plots annual year-over-year changes in WTI against U.S. CPI inflation from 2000 to 2024. The relationship is strong and positive (r = +0.65). The 2022 observation, oil up nearly 39% year over year and headline inflation at 8.0%, is a clear outlier on the upper right, consistent with Presno and Prestipino's claim that the Russia-Ukraine oil shock contributed roughly one percentage point to headline inflation on impact. The 2009 and 2015 observations, when oil prices collapsed, lie at the lower left where the OLS line predicts they should.

Figure 2. Testing AS-AD: Oil Price Shocks vs. U.S. Inflation (Annual, 2000-2024) — r = +0.65



Source: WTI from Yahoo Finance (annual averages), CPI inflation from World Bank (FP.CPI.TOTL.ZG). Outlier years labeled.

Figure 2. WTI year-over-year change vs. U.S. CPI inflation, annual 2000–2024. OLS fit shown in red.

Key finding - AS-AD

A 10% increase in WTI is associated with roughly a 0.4 percentage point rise in headline inflation in the same year. This is consistent with both Dornbusch, Fischer, and Startz (Chapter 5) and the FEDS Notes DSGE estimates. The textbook supply-shock prediction holds.

4.3 Crowding Out: Where the Textbook Goes Wrong

Figure 3 plots U.S. federal debt as a share of GDP against the 10-year Treasury yield, with points colored by year. The textbook loanable-funds prediction is that as debt rises, demand for loanable funds rises and the equilibrium real interest rate must rise to clear the market. The data show the opposite. The correlation is −0.78. As federal debt climbed from 33% of GDP in 2000 to 118% in 2024, the 10-year yield generally fell rather than rose.

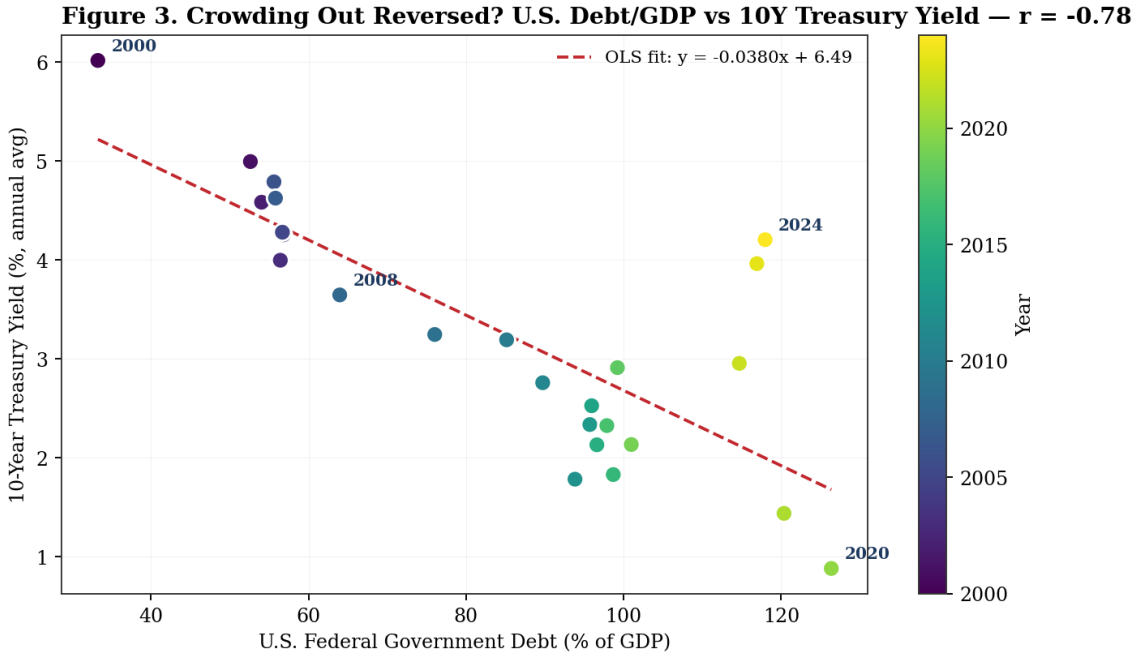
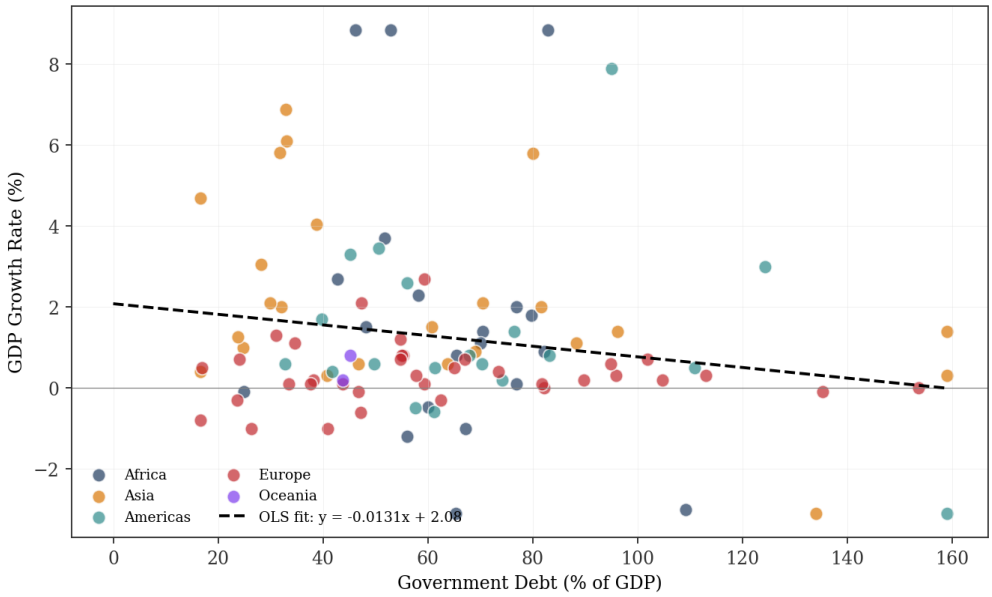


Figure 3. U.S. federal debt-to-GDP vs. 10-year Treasury yield, 2000–2024. Colors denote year. The downward-sloping fit contradicts the textbook crowding-out prediction.

Why? The textbook model assumes the saving supply curve and the central bank's behavior are independent of the demand for loanable funds. Neither is true in modern U.S. data. The largest run-ups in federal debt, 2008–2010 and 2020, coincided with deep recessions during which (a) household and corporate saving rose sharply, shifting the supply curve right, and (b) the Federal Reserve cut its policy rate to the zero lower bound and undertook large-scale asset purchases that compressed the long end of the yield curve. The result is that debt and interest rates moved in opposite directions throughout the period.

This does not mean crowding out is absent, it means the channel is different than the textbook suggests. Figure 4 brings in cross-country evidence to identify the real mechanism. Plotting government debt against GDP growth for 101 countries (after dropping missing values and winsorizing), the relationship is negative but modest ($r = -0.19$). High-debt countries grow more slowly. But the cross-country correlation between debt and interest rates is essentially zero (+0.02). The crowding out that exists in the world operates not through interest rates but through slower long-run growth, exactly what Reinhart and Rogoff (and others) have documented.

Figure 4. Cross-Country Evidence: Debt/GDP vs Growth (173 Countries) — r = -0.19



Source: World Economics cross-country snapshot dataset (n=101 after dropping missing values, winsorized at 2.5/97.5%).

Figure 4. Cross-country relationship between government debt-to-GDP and GDP growth, 173 countries (n = 101 after dropping missing values). Color denotes region.

Key finding - Crowding out

The textbook crowding-out story is right about the destination (slower growth in high-debt economies) but wrong about the mechanism (rising real interest rates). In modern data, counter-cyclical monetary policy and high private saving during recessions suppress the rate channel. The drag on growth appears to operate through other channels, fiscal space, expectations, and productivity, not the loanable funds market.

Table 2. Summary of Hypothesis Tests

#	Hypothesis	Model	r	Result
H1	Oil ↑ → Inflation ↑	*AS-AD supply shock*	**+0.65**	Strongly supported
H2	Inflation ↑ → 10Y Treasury ↑	*IS-LM / Fisher effect*	**+0.23**	Weak / partial
H3	Debt/GDP ↑ → Investment/GDP ↓	*Loanable funds*	**−0.29**	Moderate
H4	Debt/GDP ↑ → 10Y Treasury ↑	*Crowding-out channel*	**−0.78**	**REVERSED**
H5	Oil shock (monthly) ↔ S&P 500 (12-mo lead)	*IS-LM transmission*	**−0.35**	Confirmed (with lag)
H6	Cross-country: Debt ↑ → Growth ↓	*Long-run crowding out*	**−0.19**	Modest, via growth

Note: Pearson correlations. n = 25 for U.S. annual, n = 310 for monthly, n = 101 for cross-country.

5. Discussion and Conclusion

The empirical picture is partly consistent with the FEDS Notes article and partly at odds with the standard textbook framework. Presno and Prestipino's central finding, that an oil supply shock raises U.S. headline inflation noticeably while leaving core inflation and GDP largely intact, is supported by the U.S. annual data assembled here. The AS-AD prediction holds. The IS-LM rate channel is partially visible: long-term yields do tend to rise during oil shock episodes,

but the contemporaneous correlation is weak in annual data, likely because the Fed's reaction function shifted over the sample period and the LM curve was effectively flat at the zero lower bound for much of 2008–2021.

The crowding-out finding is the most consequential. The data does not support the textbook loanable-funds prediction that rising public debt drives up interest rates. They do offer modest support for the broader claim that high-debt economies experience slower long-run growth. The implication is that the loanable funds market, drawn cleanly in Chapter 11 of Dornbusch, Fischer, and Startz, is too clean. Saving supply is not exogenous to the business cycle. Monetary policy is not a passive observer of fiscal choices. In a world with active central banks and counter-cyclical private saving, deficits and rates can comfortably move in opposite directions for decades at a time, even if the long-run growth penalty of high debt is real.

For the current 2025–2026 oil shock, the framework predicts a modest, transitory rise in headline inflation, a contained reaction in core inflation and real activity, and a tightening of long-term rates that propagates with a lag to equity valuations. The S&P 500 and the 10-year Treasury yield have both risen alongside oil, which is consistent with the contemporaneous patterns identified above. Whether the 12-month lagged drag on equities materializes will depend importantly on the path of monetary policy and on whether the oil shock proves persistent. The DSGE-style modesty of the effects estimated by Presno and Prestipino is a reasonable baseline expectation.

Three limitations bear noting. First, all U.S. macroeconomic series here are annual; finer-frequency tests would identify shock-response dynamics with more precision and are worth pursuing in future work. Second, the cross-country dataset is a single-period snapshot, which does not separate within-country variation from cross-country differences in institutions, monetary regimes, and currency status. Third, the analysis is correlational. Cleanly identifying the causal effect of oil shocks would require an instrument or structural framework like the DSGE in Presno and Prestipino. Within those limits, the empirical picture is clear: the AS-AD framework remains a useful workhorse for thinking about oil shocks, and the standard crowding-out mechanism needs to be modernized to incorporate the role of active monetary policy and endogenous private saving.

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